

ex 14

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Run

Save

Output

Input

```
1 clear;
2 clc;
3 close all;
4
5 %% PARAMETERS
6
7 num_vehicles = 10;      % Number of vehicles
8 num_slots = 20;        % Number of time slots/messages
9
10 % Latency parameters: mean and standard deviation (ms)
11 %      Mean      Std
12 latency_params = [1.0  0.2;      % DSRC
13                   2.0  0.5;      % C-V2X
14                   0.5  0.1];    % 5G NR V2X
15
16 tech = {'DSRC','C-V2X','5G NR V2X'};
17
18 latency_all = zeros(num_vehicles, ...
19                    num_slots, ...
20                    length(tech));
21
22 %% =====
23 % SIMULATE LATENCY
24 % =====
25
26 for t = 1:length(tech)
27
28     mean_delay = latency_params(t,1);
29     std_delay = latency_params(t,2);
30
31     for v = 1:num_vehicles
32
33         for s = 1:num_slots
34
35             % Propagation delay
36             prop_delay = mean_delay + ...
37                 std_delay * randn;
```

```
=====
V2X LATENCY COMPARISON
=====
DSRC : 1.29 ms
C-V2X : 2.15 ms
5G NR V2X : 0.78 ms
=====
```

[Execution complete with exit code 0]

